transport and highways. In the light of these incidents, the Centre is also planning to issue quality-centric guidelines for electric vehicles (EVs) in the country.

In March, the video footage of a burning Ola Electric scooter in Pune had gone viral on social media. Almost a month later, on April 24, Ola said it was recalling 1,441 units of its e-scooters after conducting a "detailed diagnostic and health check of the scooters in that specific batch". Bhavish Aggarwal, the company's co-founder and CEO, said on Twitter that world-class agencies have been roped in to look into the cases. In another development, Okinawa Autotech, whose Praise brand of scooters had been involved in the Trichy fire incident, has voluntarily recalled 3,215 of its Praise Pro electric scooters. Pure EV, another scooter maker, announced that it was recalling 2,000 of its vehicles after two of them caught fire in two separate cases reported from Nizamabad in Telangana (where one person was killed) and Chennai in Tamil NaduThe Centre has asked the Centre for Fire Explosive and Environment Safety (CFEES) to probe the incidents and suggest remedial steps to prevent such incidents in the future.

But what does this mean for India's nascent electric two-wheelers industry that has been growing at a robust pace? Sales of e-scooters rose 132 per cent in the calendar year 2021 to 233,971 units compared to a year ago, according to industry reports. But the recent events could play spoilsport. "Such incidents will impact the buyer's safety perception of EVs," says a Delhi-based consultant who did not wish to be named. If not addressed, the problem can even impact the electric cars market. "Where a two-wheeler is concerned, a user can perhaps run away. In a car, you are trapped in a cabin, so the risk is higher," he says.

Although no cases have been reported in electric cars so far in India, one clear area of concern seems to be the batteries of EVs. According to media reports, lithium cells, particularly the lithium-NMC (nickel-manganese-cobalt) ones that are primarily used in EVs, have a thermal stability issue. That is, if the cell cannot cool down sufficiently, it could suffer from 'thermal runaway'. According to an article in TechRadar, a portal that gives tech-buying advice, though lithium batteries have a number of advantages that make them ideal for e-bikes—they can be charged and discharged hundreds of times, are relatively lightweight, compact and less toxic—they are also prone to fires since the electrolyte fluid between the two electrodes within the battery is highly combustible. When the battery cells overheat, the heat and pressure rise to a point where the battery explodes.

Some experts have also questioned the large-scale use of batteries made in China. "Low-cost batteries are flowing in from China. China clearly differentiates between what it sells to developed nations like the US and what it exports to emerging markets like India," alleges the auto consultant. Apart from the highly inflammable nature of lithium-ion, or Li-ion, batteries, another disadvantage is that they are quite expensive and contribute to 20-40 per cent of an EV's cost. That brings us to the question whether there are alternatives to lithium-ion batteries. One promising alternative is the sodium-ion, or Na-ion, batteries. They are said to be significantly cheaper than Li-ion ones. According to an article in the *New Scientist*, there is research being carried out on magnesium, sea water, glass batteries (where the electrolyte is made of glass spiked with sodium ions), fuel cells and liquid batteries (where all components are dissolved in liquids) in the quest for alternatives to Li-ion batteries.

However, these innovations will take time to be commercialised and, until then, Li-ion batteries remain the best option. Experts say it is imperative that companies do a thorough enquiry on the reasons for the fire incidents and resort to recalling batches of vehicles that are prone to fire. The e-scootermakers also need to conduct proper validation of the batteries—a process of quality control—to make them more secure. The process might be expensive, but auto makers are left with few options at this point. On April 20, the Niti Aayog came out with a draft policy on battery-swapping for EVs. Under this, users can purchase an EV without the battery, which lowers the vehicle cost. Batteries will be provided for a fee by service providers, who will have to abide by a strict testing protocol for better battery safety. The government and the industry need to work together if they wish to regain customer confidence in e-bikes. ■

A BURNING ISSUE

THE PROBLEMS

- Half a dozen incidents of electric bikes catching fire in India have been reported in less than a month. The problems seem to be with lithium-ion (Li-ion) batteries.
- The electrolyte fluid between the two electrodes in Li-ion batteries is highly combustible. When the battery cells overheat, the heat and pressure rise to a point where the battery explodes.

THE SOLUTION

- Sodium-ion or Na-ion batteries are said to be promising alternatives. They are also cheaper than their Li-ion counterparts.
- Research is being carried out on magnesium, sea water, glass batteries, fuel cells and liquid batteries as well

SPEED-BREAKER?

233,971: Total number of units of e-scooters sold in India in calendar year 2021; **132%** higher than the previous year. The incidents of fire, however, could scuttle this growth.

APART FROM BEING HIGHLY INFLAMMABLE, LI-ION BATTERIES ARE ALSO EXPENSIVE AND CONTRIBUTE TO 20-40 PER CENT OF AN EV'S COST